Sean Rhee

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Education

New York University, Ph.D. Computer Science *Sept. 2025 ~ Present*

Northwestern University, B.S. Computer Science *Sept. 2021 ~ Jun. 2025*

Experience

Graduate Researcher, New York University *Sept. 2025 ~ Present*

- Working on a generic IR for programmable networks.
- Developing formally verified ecosystem for network program reasoning to prove program equivalence and transformation correctness.

Research Internship, Argonne National Laboratory *May 2025 ~ July 2025*

- Introduced circulant graph algorithms into MPICH.
- Tuned broadcast implementation, observing strict improvements over existing algorithms across message and cluster sizes.

Undergraduate Researcher, Northwestern University

Parameterized Collective Communication Library *Mar. 2023 ~ Oct. 2025*

- Developed parameterized collective communication algorithms for GPUs.
- Observed 35-110% speedup over state-of-the-art algorithms.
- Wrote the CUDA kernels to implement custom algorithms. Built profiling tools to measure network properties and collective operation performance.

LLMSpec + TRACTOR *Mar. 2024 ~ Dec. 2024*

- Explored lifting large scientific codebases from C/C++/Fortran into a high level specification language for property testing and portability.
- Built code analysis and translation pipelines to convert C/C++/CUDA functions and kernels into idiomatic Python representations.
- Applied similar techniques for macro translation in the context of DARPA's Translating All C to Rust (TRACTOR) initiative.

Publications

[1] Liu, Peizhi, Sean Rhee, Michael Wilkins, and Peter Dinda. "Parameterized Algorithms and Parameter Selection for Fast GPU-GPU Collective Communication." In *2025 33rd International Symposium on Modeling, Analysis and Simulation of Computer and Telecommunication Systems (MASCOTS)*, pp. 1-9. IEEE, 2025.

Instruction

Undergraduate Teaching Assistant, Northwestern University

Computer Systems *w/ Branden Ghena, Winter 2025*
Computer Systems *w/ Peter Dinda, Fall 2024*
Systems Security *w/ Yan Chen, Winter 2024*
Generative Art *w/ Kate Compton, Fall 2023*
Computer Systems *w/ Nikos Hardavellas, Spring 2023*
Computer Systems *w/ Branden Ghena, Winter 2023*

Honors & Awards

CRA Outstanding Undergraduate Research Awards Honorable Mention *Dec. 2024*

Peter and Adrienne Barris Outstanding Peer Mentors Award *Apr. 2024*